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IKB42603 • Cloud Security

WEEK 1 • LESSON 1

# Fundamentals of Cloud Computing & Architectural Characteristics

*Deployment models · Service models · Cloud roles · Security risks*



# What you will be able to do



## 1. Define cloud computing

Using the NIST reference model and everyday examples.

## 2. Explain the essential characteristics

And read each one through a security lens.

## 3. Differentiate deployment models

Public, private, community and hybrid.

## 4. Compare service models

SaaS, PaaS and IaaS responsibilities.

## 5. Identify key cloud roles

Provider, consumer, partner, carrier, auditor.

## 6. Recognise cloud security risks

And the shared responsibility model.

# Curriculum alignment — CLO · PLO · MQF · clusters

## CLO 1

Examine how cloud computing security concepts are applied to solve real-world technical problems.



### Programme outcome

PLO1 · Knowledge  
MQF Level i  
Bloom C3 (Apply)



### Values & skills

SC1 · System Thinking



### Sustainability (SDG)

SDG 9.1  
Resilient, inclusive infrastructure



### Assessment

Test  
Final Test

*Constructive alignment: CLO → PLO → MQF, with VBE (values), SC (skills) and SDG clusters, assessed as shown.*

# 01

## PART 1

# What is Cloud Computing?



## DEFINITION

# Cloud computing, defined

A model for delivering computing services over the internet — instead of owning and running the hardware yourself.

- Servers & compute
- Storage
- Databases
- Networking
- Software & platforms



**On-demand access + pay-as-you-go**

*Everyday examples: Google Drive · Amazon Web Services · Microsoft Azure · Netflix*

*“Cloud is simply someone else’s computer — which is exactly why security is a shared responsibility.”*

# Five essential characteristics — read through a security lens



## On-demand self-service

Provision in seconds via portal / API.

**Security:** Misconfig can happen with no gatekeeper.



## Broad network access

Reachable from any device, anywhere.

**Security:** Larger attack surface.



## Resource pooling

Shared multi-tenant infrastructure.

**Security:** Isolation & side-channel risks.



## Rapid elasticity

Auto scale up and down on demand.

**Security:** Economic Denial of Sustainability.



## Measured service

Usage metered and billed precisely.

**Security:** Telemetry powers security monitoring.

# Why do organisations move to the cloud?



## Cost efficiency

CapEx → OpEx. Pay only for what you use; no idle hardware.



## Scalability

Absorb demand spikes on-demand, then scale back down.



## Flexibility & speed

Spin up and tear down environments in minutes.



## Global reach

Deploy close to users through provider regions.

**Discussion:** Which of these drivers matters most for a Malaysian SME — and what new risk does it introduce?

# 02

## PART 2

# Architectural & Technological Influences



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# Four technologies that make the cloud possible



## Virtualization

Abstracts physical hardware into flexible virtual resources. The engine of elasticity and pooling.



## Distributed systems

Workloads span many machines and data centres for resilience and massive scale.



## Service-Oriented Architecture

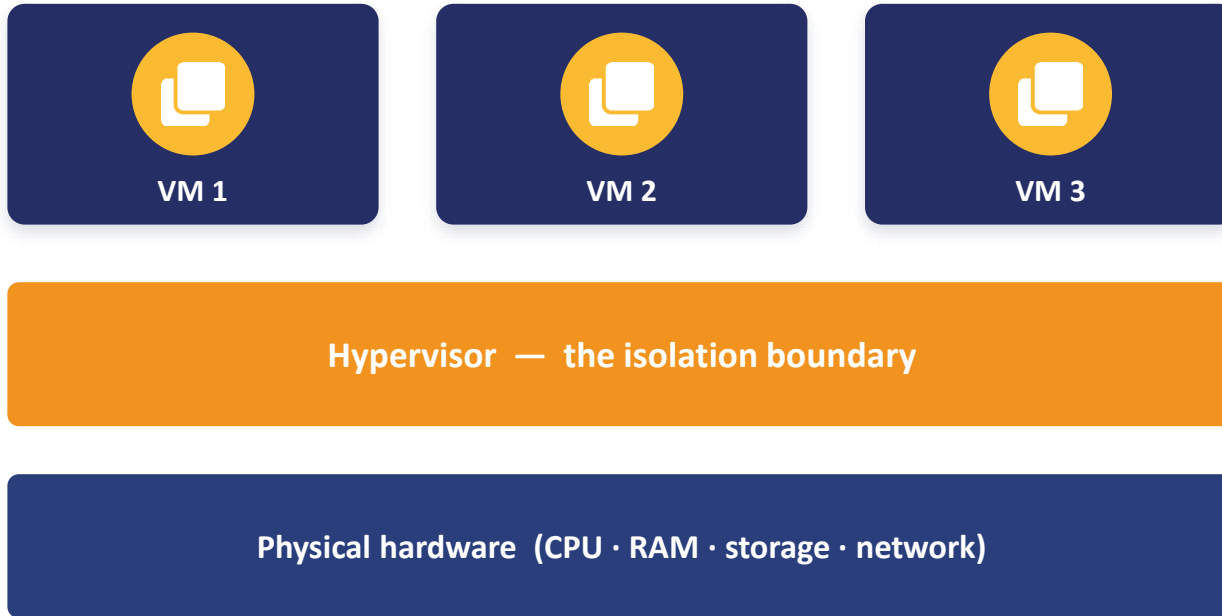
Applications split into loosely-coupled services scaled independently — the root of microservices.



## APIs & web services

Everything is controlled via APIs — powerful, but a prime attack surface.

# Virtualization: one server, many machines



## Three flavours

- **Server virtualization** — Many VMs on one host
- **Storage virtualization** — Pooled logical volumes
- **Network virtualization** — Software-defined networks

**Security tension:** *compromise the hypervisor and tenant isolation fails (Lesson 3).*

# 03

PART 3

## Cloud Deployment Models



# Four deployment models at a glance



## Public cloud

Third-party owned, shared, internet-delivered.  
Cheapest, most scalable.



## Private cloud

Dedicated to one organisation. Most control, higher cost.



## Community cloud

Shared by orgs with common goals (gov, healthcare).  
Shared cost & governance.



## Hybrid cloud

Public + private combined. Flexibility with data control.

# Public vs Private — the control / cost trade-off

## Public cloud

- Owned by third-party providers
- Accessed over the internet
- Pay-per-use pricing
- AWS · Azure · Google Cloud

### ✓ Advantages

Scalable & cost-effective · No hardware to manage

### ⚠ Trade-offs

Less control · Shared environment

## Private cloud

- Dedicated to one organisation
- On-prem or hosted
- Full administrative control
- Often driven by regulation

### ✓ Advantages

High security & isolation · Meets data-sovereignty rules

### ⚠ Trade-offs

Expensive to build & run · You carry the ops burden

# Community & Hybrid — the collaborative and the blended



## Community cloud

*Shared by organisations with common goals or regulations.*

- Example: government agencies, healthcare consortium
- Benefit: shared cost & common standards
- Challenge: governance & liability complexity



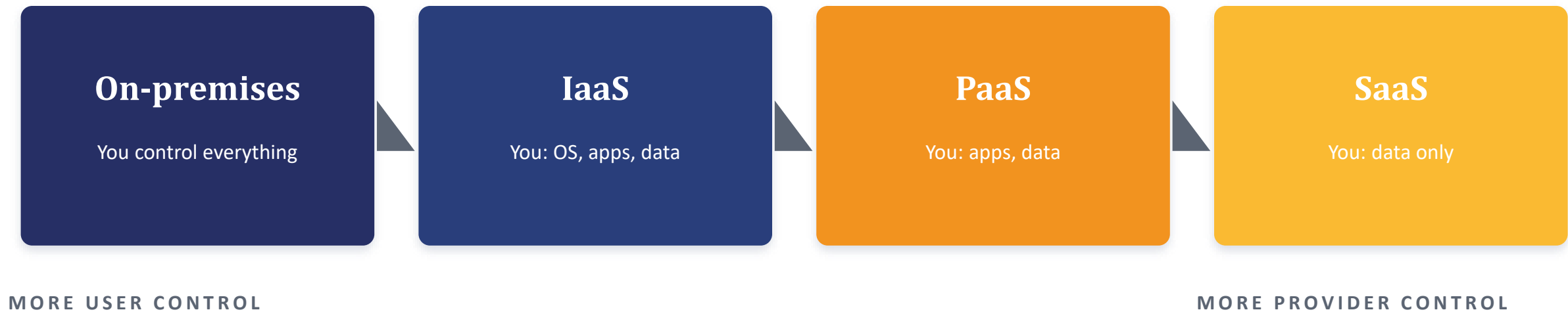
## Hybrid cloud

*A deliberate blend of private + public, connected together.*

- Sensitive data → private tier
- General / bursty apps → public tier
- Challenge: the connecting seam must be secured

KEY IDEA

## Scope of control: user → provider



**Rule of thumb:** *whoever controls a layer must secure it — the gap in between is where breaches happen.*

# 04

PART 4

## Cloud Service Models



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# SaaS · PaaS · IaaS — who manages what

**SaaS**

**Finished software, used in a browser**

**You manage: data & access**

**Examples:** *Gmail · Dropbox · Salesforce*

**PaaS**

**A platform to deploy your code onto**

**You manage: apps & data**

**Examples:** *Google App Engine · Heroku*

**IaaS**

**Raw infrastructure to build on**

**You manage: OS, apps & data**

**Examples:** *AWS EC2 · Azure VMs*

# Software as a Service (SaaS)



Fully managed by the provider and accessed through a browser — nothing to install or patch.

## Responsibility split

- Provider manages: infrastructure, platform, application
- You manage: your data and who can access it

## Headline security risk

**Data privacy & account access**

**Examples:** *Gmail · Microsoft 365 · Dropbox · Salesforce*

# Platform as a Service (PaaS)



A ready-made platform for developers to build and deploy applications without managing servers.

## Responsibility split

- Provider manages: OS, runtime, infrastructure
- You manage: your application code and data

## Headline security risk

**Platform dependency & code security**

**Examples:** *Google App Engine · Heroku · Azure App Service*

# Infrastructure as a Service (IaaS)



Provides virtual machines, storage and networks — the most flexible and most hands-on model.

## Responsibility split

- Provider manages: physical hardware, virtualization
- You manage: OS, apps, data, network config

## Headline security risk

**Misconfiguration & OS hardening**

**Examples:** *AWS EC2 · Azure Virtual Machines · Google Compute Engine*

## SUMMARY

# Service models compared

| Aspect           | SaaS              | PaaS                 | IaaS                      |
|------------------|-------------------|----------------------|---------------------------|
| What you get     | Finished software | Development platform | Raw infrastructure        |
| You manage       | Data & access     | Apps + data          | OS + apps + data          |
| Provider manages | Almost everything | Runtime & below      | Hardware & virtualization |
| Flexibility      | Lowest            | Medium               | Highest                   |
| Headline risk    | Data privacy      | Platform dependency  | Misconfiguration          |
| Example          | Gmail, Salesforce | App Engine, Heroku   | AWS EC2, Azure VM         |

# 05

PART 5

## Cloud Roles & Responsibilities



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# Who's who in the cloud



## Cloud provider

Builds & delivers the services.



## Cloud consumer

Organisation or person using them.



## Partner / broker

Aggregates, customises & manages services.



## Cloud carrier

Provides the network & connectivity.



## Cloud auditor

Independently assesses security & compliance.



## Security owner

Accountability follows the role — not the vendor.

# The shared responsibility model



## Provider secures the cloud

- Physical data centres
- Hardware & network fabric
- Virtualization / hypervisor
- Managed service internals



## You secure what's in the cloud

- Your data & its encryption
- Identity & access management
- Configuration & policies
- Guest OS & applications (IaaS)

**Remember:** *the provider secures the cloud; you secure what you put in it. Most breaches live on the right-hand side.*



# 06

PART 6

## Risks, Best Practices & Case Study



# Top cloud security risks



## Data breaches

Unauthorised access to sensitive data.



## Data loss

Permanent loss via deletion, corruption or provider failure.



## Account hijacking

Stolen credentials → legitimate-looking access.



## Insecure APIs

Everything is API-driven — weak APIs are a direct way in.



## Insider threats

Malicious or careless staff, customer or provider side.



## Misconfiguration

The #1 real-world cause of cloud incidents.

# Best practices — the essentials



## Strong authentication

MFA defeats most credential attacks. (Lesson 5)



## Data encryption

In transit and at rest. (Lesson 4)



## Regular, tested backups

Defend against data loss & ransomware.



## Least-privilege access

Limit blast radius of a breach. (Lessons 5 & 7)



## Security monitoring

Detect & respond early. (Lesson 6)



## Discipline in the basics

Prevents the majority of incidents.

## Mini case study — the migration breach



A company migrates its customer database to public IaaS. To move fast, the team leaves a storage bucket public and reuses admin passwords with no MFA. Weeks later, the data is found openly on the internet.

### What went wrong?

Misconfiguration · weak authentication · no monitoring

### Who is responsible?

The customer — every failure sat on their side of the shared responsibility line

## Key takeaways



1 Cloud = on-demand, pay-as-you-go computing over the internet, with five essential characteristics.



2 Deployment models differ by who owns & uses the cloud — trading control against cost.



3 Service models (IaaS/PaaS/SaaS) differ by how much you manage vs the provider.



4 Security is a shared responsibility — and most breaches sit on the customer's side.



# Discussion & Q&A

- Would you trust a public cloud with your organisation's most sensitive data? Why or why not? Which best practice would you implement first — and what would it cost you to skip it?

*Next week — Lesson 2: Security Design & Architecture for Cloud Computing (CSA · NIST · ENISA)*